(d) Electricity can be used to break down aqueous or molten ionic compounds.

(i) Name the process which uses electricity to break down aqueous or molten ionic compounds.

.... [1]

(ii) Explain why the ionic compound needs to be aqueous or molten.

.... [1]

(e) Brine is concentrated aqueous sodium chloride.

(i) Name **three** substances which are manufactured by passing electricity through brine.

1 ....

2 ....

3 ....
[3]

(ii) Name a different substance formed when molten sodium chloride is used instead of concentrated aqueous sodium chloride.

.... [1]

[Total: 13]

5 This question is about copper and its compounds.

(a) Describe the bonding in a metallic element such as copper.

You may include a diagram as part of your answer.

....

....

.... [3]

(b) A metal spoon is electroplated with copper.

State what is used as:

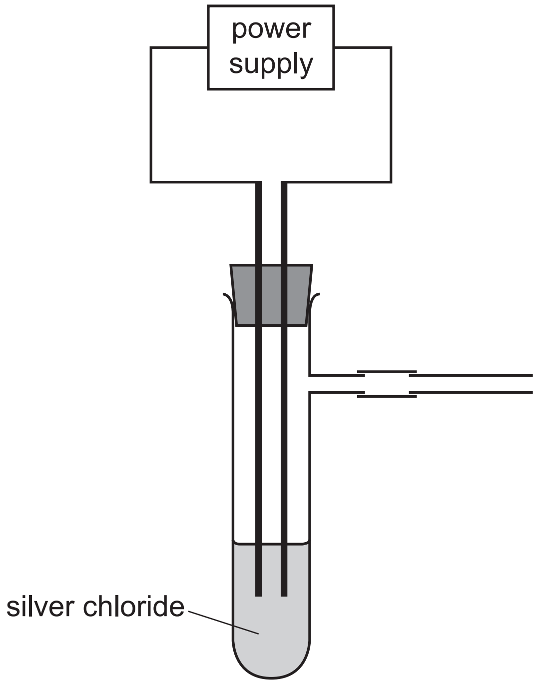
the positive electrode (anode) ....

the negative electrode (cathode) ....

the electrolyte. ....
[3]

1 Silver chloride is an ionic compound and is insoluble in water. Molten silver chloride breaks down during electrolysis. The products are chlorine and silver.
Chlorine gas is soluble in water and toxic.

A student suggests using the apparatus shown to break down silver chloride.



(a) Draw an arrow on the diagram to show where heat must be applied so that the silver chloride can break down. [1]

(b) Complete the diagram to show how chlorine gas can be collected and the volume of the chlorine measured. Label any apparatus you have drawn. [2]

(c) Give **two** observations that are made as the silver chloride breaks down.

1 ....

2 ....
[2]

(d) The person doing the experiment followed all normal laboratory safety rules.

State **one** additional safety precaution that should be taken when doing this experiment. Give a reason for your answer.

safety precaution ....

reason ....
[2]

(e) Suggest **one** reason why zinc is **not** a suitable material to use as the electrodes.

....

.... [1]

(f) The chlorine gas was bubbled into an aqueous solution of a sodium salt. The colour of the solution changed from colourless to orange.

Identify the sodium salt and explain what has happened to cause the colour change.

sodium salt ....

explanation ....

....
[2]

[Total: 10]

3 Potassium reacts with chlorine to form potassium chloride, KC*l*.

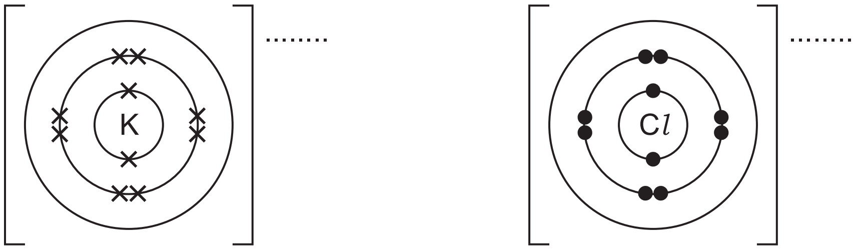
(a) Write a chemical equation for this reaction.

.... [2]

(b) Potassium chloride is an ionic compound.

Complete the diagram to show the electron arrangement in the outer shells of the ions present in potassium chloride.

Give the charges on both ions.



[3]

(c) Molten potassium chloride undergoes electrolysis.

(i) State what is meant by the term *electrolysis*.

....

.... [2]

(ii) Name the products formed at the positive electrode (anode) and negative electrode (cathode) when molten potassium chloride undergoes electrolysis.

anode ....

cathode ....
[2]

(d) Concentrated aqueous potassium chloride undergoes electrolysis.

(i) Write an ionic half-equation for the reaction at the negative electrode (cathode).

.... [2]

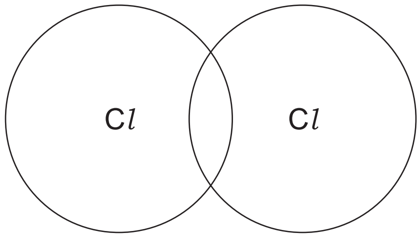
(ii) Name the product formed at the positive electrode (anode).

.... [1]

(iii) Name the potassium compound that remains in the solution after electrolysis.

.... [1]

(e) Complete the dot-and-cross diagram to show the electron arrangement in a molecule of chlorine, Cl_2.
Show the outer electrons only.



[1]

(f) The melting points and boiling points of chlorine and potassium chloride are shown.

	melting point / °C	boiling point / °C
chlorine	–101	–35
potassium chloride	770	1500

(i) Deduce the physical state of chlorine at –75 °C. Use the data in the table to explain your answer.

physical state ....

explanation ....

....
[2]

(ii) Explain, in terms of structure and bonding, why potassium chloride has a much higher melting point than chlorine.

Your answer should refer to the:
- types of particle held together by the forces of attraction
- types of forces of attraction between particles
- relative strength of the forces of attraction.

....

....

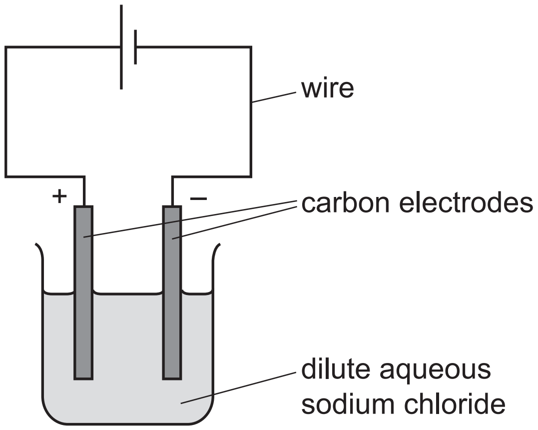
....

....

.... [3]

[Total: 19]

4 A student carries out an electrolysis experiment using the apparatus shown.



The student uses dilute aqueous sodium chloride.

(a) State the name given to any solution which undergoes electrolysis.

.... [1]

(b) Hydroxide ions are discharged at the anode.

(i) Complete the ionic half-equation for this reaction.

..........$OH^-(aq) \rightarrow$ .... + $O_2(g)$ + $4e^-$ [2]

(ii) Explain how the ionic half-equation shows the hydroxide ions are being oxidised.

.... [1]

(c) Describe what the student observes at the cathode.

.... [1]

(d) Write the ionic half-equation for the reaction at the cathode.

.... [2]

(e) The student repeats the experiment using concentrated aqueous sodium chloride.

(i) Describe what the student observes at:

- the cathode ....
- the anode. ....

[2]

(ii) The student added litmus to the solution after the electrolysis of concentrated aqueous sodium chloride.

State the colour seen in the solution. Give a reason for your answer.

colour of solution ....

reason ....

[2]

(f) Carbon electrodes are used because they are inert.

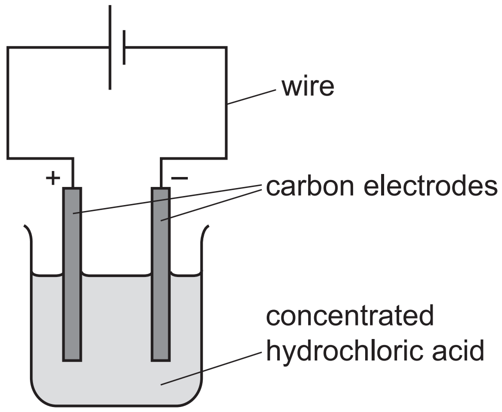
State another element that can be used instead of carbon.

.... [1]

[Total: 12]

3 This question is about electrolysis.

Concentrated hydrochloric acid is electrolysed using the apparatus shown.



(a) Chloride ions are discharged at the anode.

(i) Complete the ionic half-equation for this reaction.

..........$Cl^-(aq) \rightarrow$ ....(g) + ...e^- [2]

(ii) State whether oxidation or reduction takes place. Explain your answer.

....

.... [1]

(b) Describe what is seen at the cathode.

.... [1]

(c) Write the ionic half-equation for the reaction at the cathode.

.... [2]

(d) The pH of the electrolyte is measured throughout the experiment.

(i) Suggest the pH of the electrolyte at the beginning of the experiment.

.... [1]

(ii) State how the pH changes, if at all, during the experiment.

Explain your answer.

....

.... [2]

(e) The electrolysis is repeated using molten lead(II) bromide.

Describe what is seen at the:

- cathode ....
- anode. ....

[2]

(f) State **two** properties of graphite (carbon) which make it suitable for use as an electrode.

1 ....

2 ....

[2]

[Total: 13]

2 This question is about electrolysis.

(a) State the meaning of the term *electrolyte*.

....

.... [2]

(b) The table gives information about the electrolysis of two electrolytes. Carbon (graphite) electrodes are used in each experiment.

(i) Complete the table to show the observations and products of electrolysis.

	positive electrode (anode)		negative electrode (cathode)	
electrolyte	observations	name of product	observations	name of product
aqueous copper(II) sulfate	colourless bubbles			
concentrated aqueous sodium bromide			colourless bubbles	hydrogen

[5]

(ii) Hydrogen is produced at the negative electrode (cathode) during the electrolysis of concentrated aqueous sodium bromide.

Write the ionic half-equation for this reaction.

.... [2]

(iii) State **two** reasons why carbon (graphite) is suitable to use as an electrode.

1 ....

2 ....
[2]

(iv) Name the particle responsible for the conduction of electricity in the metal wires used in a circuit.

.... [1]

[Total: 12]

5 Electrolysis of concentrated aqueous sodium chloride using inert electrodes forms chlorine, hydrogen and sodium hydroxide.

(a) What is meant by the term *electrolysis*?

....

....

.... [2]

(b) Name a substance that can be used as the inert electrodes.

.... [1]

(c) Write an ionic half-equation for the formation of hydrogen during this electrolysis.

.... [1]

(d) Give the formulae of the **four** ions present in concentrated aqueous sodium chloride.

.... [2]

(e) Explain how sodium hydroxide is formed during this electrolysis.

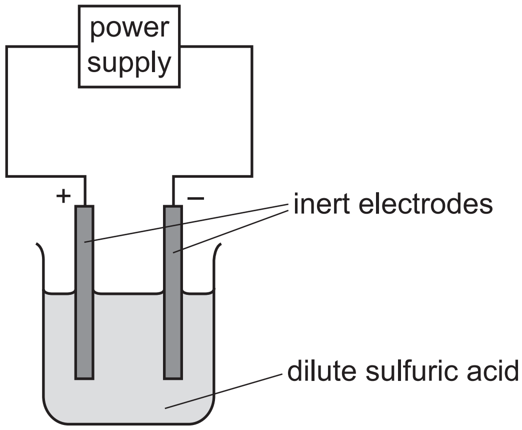
....

....

.... [2]

[Total: 8]

5 **(a)** Dilute sulfuric acid is electrolysed using the apparatus shown in the diagram.



(i) State what is meant by the term *electrolysis*.

....

....

.... [2]

(ii) Explain why inert electrodes are used.

....

.... [1]

(iii) Name the products formed at each electrode.

negative electrode ....

positive electrode ....
[2]

(iv) Write an ionic half-equation for the reaction at the negative electrode.

.... [2]

5 Group I elements, Group VII elements and transition elements are found in different parts of the Periodic Table.

(a) Describe the trend in the reactivity of Group I elements.

....

.... [1]

(b) When potassium is added to water a chemical reaction occurs.

(i) State **two** observations that can be made when potassium is added to water.

....

.... [2]

(ii) Write a chemical equation for the reaction of potassium with water.

.... [2]

(c) Excess aqueous potassium iodide is added to chlorine.

(i) Write a chemical equation for the reaction that occurs when aqueous potassium iodide is added to chlorine.

.... [2]

(ii) State the final colour of the reaction mixture.

.... [1]

(d) Sodium is extracted from sodium chloride by electrolysis.

(i) State the meaning of the term *electrolysis*.

....

.... [2]

(ii) State what must be done to sodium chloride before it can be electrolysed to produce sodium.

.... [1]

(iii) Write an ionic half-equation for the change that occurs at the cathode during this electrolysis.

.... [1]

(e) Chromium is a transition element.

- Chromium has a high melting point.
- Chromium is a good conductor of electricity.
- Many chromium compounds are soluble in water.
- Hydrated chromium(III) sulfate is green.
- Chromium forms the chlorides $CrCl_2$ and $CrCl_3$.
- Oxides of chromium act as catalysts in the manufacture of poly(ethene).

(i) Use this information to give **two** properties of chromium which are different from properties of Group I elements such as sodium.

1 ....

2 ....
[2]

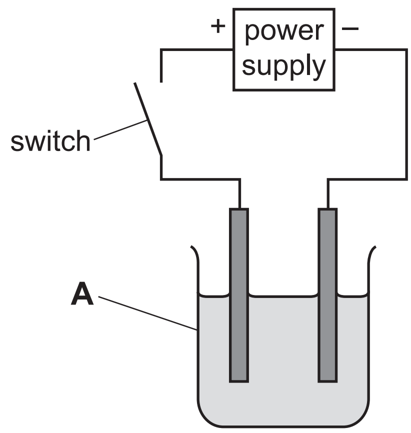
(ii) Use this information to give **two** properties of chromium which are similar to properties of Group I elements such as sodium.

1 ....

2 ....
[2]

[Total: 16]

1 The diagram shows the apparatus used to pass an electric current through concentrated hydrochloric acid. Hydrogen and chlorine were formed at the electrodes.



(a) Name the item of apparatus labelled **A**.

.... [1]

(b) The electrodes were made of platinum.

(i) Give **two** reasons why platinum is a suitable material for the electrodes.

1 ....

2 ....
[2]

(ii) Suggest another material suitable to use as electrodes in this experiment.

.... [1]

(c) The teacher doing this experiment wore safety glasses, gloves, had their hair tied back and stood up throughout the experiment.

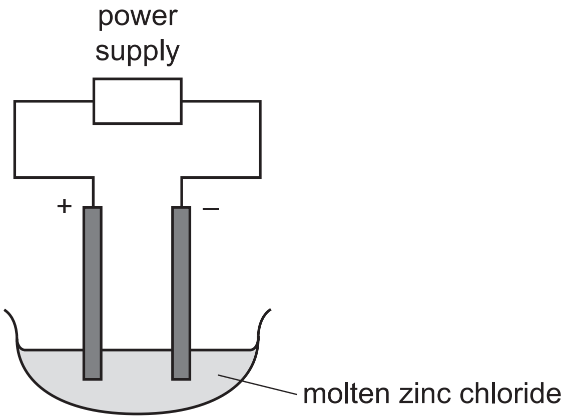
State **one** other safety precaution that should be taken when doing this experiment.
Explain your answer.

safety precaution ....

explanation ....
[2]

[Total: 6]

1 A chemist heated solid zinc chloride until it became molten. The apparatus shown was then used to pass electricity through the molten zinc chloride using inert electrodes.



A silver-coloured solid was formed at the negative electrode (cathode).

(a) Name the process of breaking down a substance using electricity.

.... [1]

(b) A Bunsen burner was used to heat the zinc chloride.

Describe how a Bunsen burner is adjusted to give a very hot flame.

....

.... [1]

(c) Suggest and explain the expected observation at the positive electrode (anode).

....

.... [2]

(d) Suggest why iron electrodes **cannot** be used in this experiment.

.... [1]

(e) (i) What difference would the chemist observe at the negative electrode if aqueous zinc chloride were used, rather than molten zinc chloride?
Explain you answer.

difference ....

explanation ....

....
[2]

(ii) When electricity is used to break down concentrated aqueous zinc chloride, chlorine is produced at the positive electrode.

Describe a test for chlorine.

test ....

observations ....
[2]

(f) The bottle of zinc chloride is labelled *corrosive*.

State **one** safety precaution that should be taken when using zinc chloride.

.... [1]

[Total: 10]

(d) Nickel is a transition element. Nickel is stronger than sodium.

Describe **two** other differences in the physical properties of nickel and sodium.

1 ....

2 ....
[2]

(e) Predict **one** difference in the appearance of aqueous solutions of nickel compounds compared to aqueous solutions of sodium compounds.

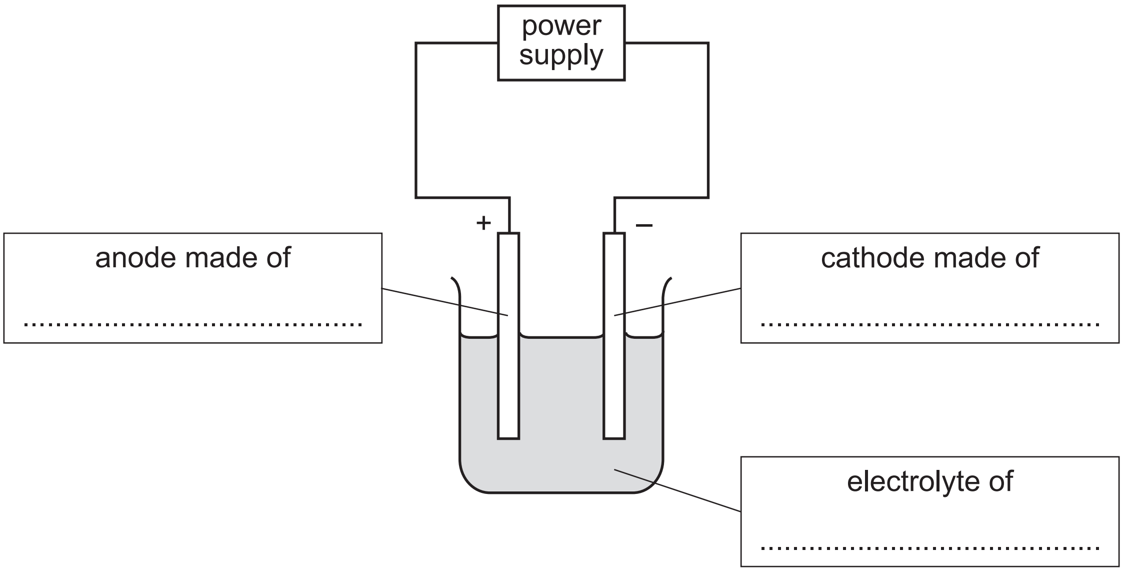
....

.... [1]

(f) Copper is refined (purified) by electrolysis. Nickel can be refined using a similar method.

(i) The diagram shows the refining of nickel by electrolysis.

Complete the labels in the boxes.



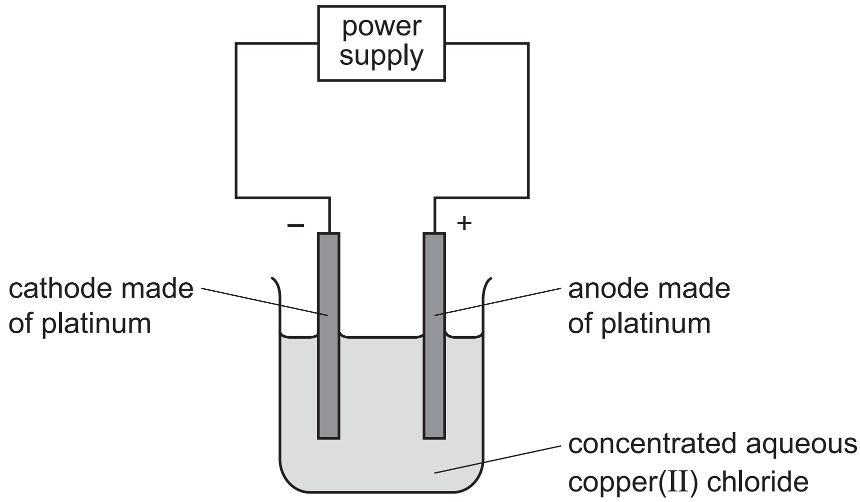
[3]

(ii) Indicate, by writing **N** on the diagram, where nickel is produced. [1]

[Total: 13]

4 Solutions of ionic compounds can be broken down by electrolysis.

(a) Concentrated aqueous copper(II) chloride was electrolysed using the apparatus shown.



The ionic half-equations for the reactions at the electrodes are shown.

negative electrode: $Cu^{2+}(aq) + 2e^{-} \rightarrow Cu(s)$

positive electrode: $2Cl^{-}(aq) \rightarrow Cl_2(g) + 2e^{-}$

(i) Platinum is a solid which is a good conductor of electricity.

State **one** other property of platinum which makes it suitable for use as electrodes.

....

.... [1]

(ii) State what would be **seen** at the positive electrode during this electrolysis.

....

.... [1]

(iii) State and explain what would happen to the mass of the negative electrode during this electrolysis.

....

....

.... [2]

(iv) The concentrated aqueous copper(II) chloride electrolyte is green.

Suggest what would happen to the colour of the electrolyte during this electrolysis.
Explain your answer.

....

....

.... [2]

(v) Identify the species that is oxidised during this electrolysis.
Explain your answer.

species that is oxidised ....

explanation ....

....
[2]

(b) Metal objects can be electroplated with silver.

(i) Describe how a metal spoon can be electroplated with silver.
Include:

- what to use as the positive electrode and as the negative electrode
- what to use as the electrolyte
- an ionic half-equation to show the formation of silver.

You may include a diagram in your answer.

....

....

....

ionic half-equation ....
[4]

(ii) Give **one** reason why metal spoons are electroplated with silver.

....

.... [1]

[Total: 13]

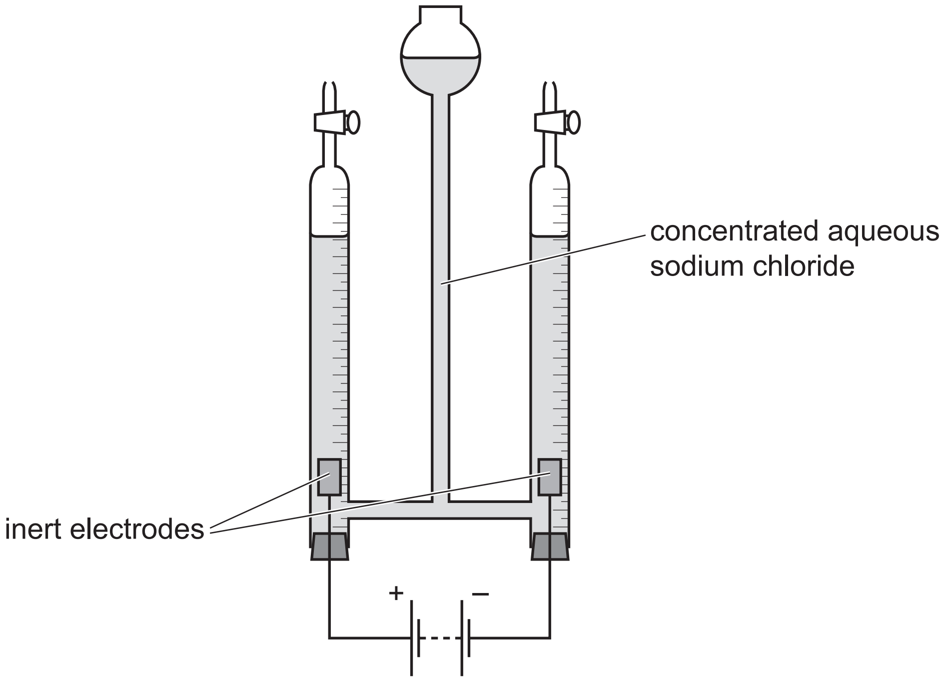
4 Many substances conduct electricity.

(a) Identify all the particles responsible for the passage of electricity in:

- graphite ....
- magnesium ribbon ....
- molten copper(II) bromide. ....

[4]

(b) A student used the following apparatus to electrolyse concentrated aqueous sodium chloride using inert electrodes.



(i) Suggest the name of a metal which could be used as the inert electrodes.

.... [1]

(ii) Name the gas formed at the positive electrode.

.... [1]

(iii) Write an ionic half-equation for the reaction occurring at the negative electrode. Include state symbols.

.... [3]

(iv) How, if at all, does the pH of the solution change during the electrolysis? Explain your answer.

....

....

.... [3]